A Codimension-One KMS Shadow Bound

A swap-walk proof and possible uses as structured base graphs

Research note

July 2026

Abstract

We give a self-contained write-up of a uniform spectral bound for a family of bipartite coset-incidence graphs attached to the affine $\widetilde{A}_2$ Kaufman–Oppenheim/Kac–Moody–Steinberg model. Let

$$R_s = \mathbb{F}_q[t]/(t^s), \qquad s \geq 2,$$

and let $G_s \leq \mathrm{SL}_3(R_s)$ be generated by

$$U_0 = \{I + atE_{31} : a \in \mathbb{F}_q\}, \quad U_1 = \{I + aE_{12} : a \in \mathbb{F}_q\}, \quad U_2 = \{I + aE_{23} : a \in \mathbb{F}_q\}.$$

For $i \in \{0,1,2\}$ put $U_{\widehat{i}} = \langle U_j : j \neq i \rangle$. We prove, using the colored swap-walk theorem for partite local spectral expanders, that the normalized bipartite coset graph

$$Y_{i,s} = \mathrm{CC}(G_s; U_i, U_{\widehat{i}})$$

has nontrivial singular value $O(q^{-1/2})$, uniformly in the congruence level s. More explicitly, in the standard colored swap-walk normalization one obtains

$$\sigma_2(Y_{i,s}) \leq 2q^{-1/2}.$$

Equivalently,

$$\left\| P_{U_i} P_{U_{\widehat{i}}} \right\|_{\ell_0^2(G_s) \to \ell_0^2(G_s)} \leq 2q^{-1/2}.$$

The point of the note is twofold. First, the proof clarifies that this codimension-one "KMS shadow" is controlled by existing local-to-global swap-walk technology; it is not a new Ramanujan-complex phenomenon by itself. Second, the same viewpoint suggests a possible route to HLMRZ-style structured base graphs: KMS coset geometries may provide highly symmetric incidence systems, but any lossless-expander application will still need non-spectral small-set density and common-neighborhood structure lemmas analogous to the cubical-complex inputs in HLMRZ.

Contents

1 Executive summary **1**

2 The affine $\widetilde{A}_2$ congruence model **2**

3 The rank-two subgroups **3**

4 The tripartite coset complex **4**

5 The local Heisenberg link calculation **5**

6 The colored swap-walk input **6**

7 Proof of the codimension-one bound **7**

8 What this result is, and what it is not **7**

9 Consequences and nearby graph shadows **8**
9.1 Layered or gallery shadows 8
9.2 Balanced chamber walks 8

10 Possible HLMRZ-style applications **9**
10.1 What HLMRZ needs from its base graphs 9
10.2 A candidate KMS replacement for cubical incidence 9
10.3 Why KMS geometry is tempting 10

11 Side observations not used in the proof **10**

12 Open directions **11**

1 Executive summary

The graph under consideration is

$$Y_{i,s} = \mathrm{CC}(G_s; U_i, U_{\widehat{i}}), \qquad U_{\widehat{i}} = \langle U_j : j \neq i \rangle.$$

It is a bipartite graph with left vertices G_s/U_i, right vertices $G_s/U_{\widehat{i}}$, and incidence given by nonempty coset intersection. Equivalently,

$$gU_i \sim hU_{\widehat{i}} \quad \Longleftrightarrow \quad gU_i \cap hU_{\widehat{i}} \neq \varnothing.$$

The normalized adjacency operator has nonzero singular values equal to those of $P_{U_i} P_{U_{\widehat{i}}}$ on $\ell^2(G_s)$, where P_H denotes right averaging over a subgroup H.

The proof has three ingredients.

1. The complementary subgroups

$$K_i := U_{\widehat{i}}$$

are Heisenberg groups of order q^3, and their pairwise intersections are exactly the root groups:

$$K_0 \cap K_1 = U_2, \qquad K_0 \cap K_2 = U_1, \qquad K_1 \cap K_2 = U_0.$$

2. The tripartite coset complex

$$X_s = \mathrm{CC}(G_s; K_0, K_1, K_2)$$

has type-i vertices G_s/K_i, chambers indexed by G_s, and cotype-i edges indexed by

$$G_s/(K_j \cap K_k) = G_s/U_i.$$

Thus $Y_{i,s}$ is precisely the incidence graph between type-i vertices and cotype-i edges in X_s.

3. Each vertex link of X_s is a Heisenberg incidence graph. Its normalized nontrivial singular value is exactly $q^{-1/2}$. The colored swap-walk theorem then gives the type-i vertex versus cotype-i edge incidence bound

$$\sigma_2(Y_{i,s}) \le 2q^{-1/2}.$$

The result is useful, but its status should be described accurately: it is a colored swap-walk consequence of local spectral expansion. The genuinely new questions begin when one asks for sharp constants, the reduced/infinite-cover spectrum, higher-rank top-new spectra, or the small-set combinatorics needed for HLMRZ-type applications.

2 The affine $\widetilde{A}_2$ congruence model

Fix a prime power q and $s \ge 2$. Put

$$R_s = \mathbb{F}_q[t]/(t^s).$$

Let $G_s \le \mathrm{SL}_3(R_s)$ be the subgroup generated by

$$\begin{aligned} U_0 &= \{I + atE_{31} : a \in \mathbb{F}_q\}, \\ U_1 &= \{I + aE_{12} : a \in \mathbb{F}_q\}, \\ U_2 &= \{I + aE_{23} : a \in \mathbb{F}_q\}. \end{aligned}$$

These are the three simple affine-root directions in the rank-three toy model. Define

$$K_i := U_{\widehat{i}} := \langle U_j : j \ne i \rangle.$$

Thus

$$K_0 = \langle U_1, U_2 \rangle, \qquad K_1 = \langle U_0, U_2 \rangle, \qquad K_2 = \langle U_0, U_1 \rangle.$$

For a subgroup $H \le G_s$, let

$$P_H f(g) = \frac{1}{|H|} \sum_{h \in H} f(gh)$$

be the right-averaging projection on $\ell^2(G_s)$, with the uniform probability measure on G_s. This is the orthogonal projection onto right-H-invariant functions.

Definition 2.1 (Coset-incidence graph)**.** For subgroups $H, K \le G$, let $\mathrm{CC}(G; H, K)$ be the bipartite graph with left vertices G/H, right vertices G/K, and incidence

$$gH \sim hK \quad \Longleftrightarrow \quad gH \cap hK \ne \varnothing.$$

Equivalently, the edges are indexed by $G/(H \cap K)$ when the natural map from $G/(H \cap K)$ to $G/H \times G/K$ is injective.

The normalized bipartite adjacency of $\mathrm{CC}(G; H, K)$ may be represented on $\ell^2(G)$ by the two projections P_H and P_K. The nonzero singular values are the singular values of

$$P_H P_K : ell^2(G) \to \ell^2(G),$$

after restricting to the appropriate invariant subspaces. The trivial singular value 1 comes from constant functions.

The main theorem is the following.

Theorem 2.2 (Codimension-one KMS shadow bound)**.** *For every prime power q, every $s \geq 2$, and every $i \in \{0,1,2\}$,*

$$\left\| P_{U_i} P_{U_{\widehat{i}}} \right\|_{\ell_0^2(G_s) \to \ell_0^2(G_s)} \leq 2q^{-1/2}.$$

Equivalently, if

$$Y_{i,s} = \mathrm{CC}(G_s; U_i, U_{\widehat{i}}),$$

then the normalized second singular value of $Y_{i,s}$ satisfies

$$\sigma_2(Y_{i,s}) \leq 2q^{-1/2}.$$

For $q \leq 4$ this bound is not numerically stronger than the trivial bound $\sigma_2 \leq 1$, but the uniform asymptotic statement is the point.

3 The rank-two subgroups

The elementary algebra of the three subgroups K_i is the backbone of the proof.

Lemma 3.1 (Heisenberg structure)**.** *The groups $K_i = U_{\widehat{i}}$ are Heisenberg groups over $\mathbb{F}_q$:*

$$\begin{aligned} K_0 &= \{I + aE_{12} + bE_{23} + cE_{13} : a, b, c \in \mathbb{F}_q\}, \\ K_1 &= \{I + aE_{23} + btE_{31} + ctE_{21} : a, b, c \in \mathbb{F}_q\}, \\ K_2 &= \{I + aE_{12} + btE_{31} + ctE_{32} : a, b, c \in \mathbb{F}_q\}. \end{aligned}$$

Moreover,

$$K_0 \cap K_1 = U_2, \qquad K_0 \cap K_2 = U_1, \qquad K_1 \cap K_2 = U_0,$$

and hence

$$K_0 \cap K_1 \cap K_2 = \{1\}.$$

Proof. For K_0, the basic commutator relation is

$$[I + aE_{12}, I + bE_{23}] = I + abE_{13}.$$

The E_{13}-root subgroup is central in the upper unitriangular subgroup generated by U_1 and U_2, giving the displayed form of K_0.

For K_1, one has

$$[I + aE_{23}, I + btE_{31}] = I + abtE_{21}.$$

The new tE_{21} direction commutes with both generating directions inside this rank-two subgroup, so

$$K_1 = \{I + aE_{23} + btE_{31} + ctE_{21} : a, b, c \in \mathbb{F}_q\}.$$

Similarly,

$$[I + btE_{31}, I + aE_{12}] = I + abtE_{32},$$

which gives the displayed form of K_2.

The supports of the three displayed matrix models show the intersections. The only common root direction of K_0 and K_1 is E_{23}, hence $K_0 \cap K_1 = U_2$. The only common root direction of K_0 and K_2 is E_{12}, hence $K_0 \cap K_2 = U_1$. The only common root direction of K_1 and K_2 is tE_{31}, hence $K_1 \cap K_2 = U_0$. Since these three root groups are distinct, the triple intersection is trivial. □

4 The tripartite coset complex

Define a tripartite two-dimensional coset complex X_s by declaring its type-i vertices to be

$$X_s(i) = G_s/K_i,$$

and its chambers to be triples

$$(gK_0, gK_1, gK_2), \qquad g \in G_s.$$

Because $K_0 \cap K_1 \cap K_2 = \{1\}$, chambers are naturally indexed by elements of G_s.

Let $\{i, j, k\} = \{0, 1, 2\}$. A cotype-$i$ edge is an edge using the two colors j and k. Such edges are indexed by

$$G_s/(K_j \cap K_k).$$

By Lemma 3.1,

$$K_j \cap K_k = U_i.$$

Therefore the cotype-i edges are indexed by

$$G_s/U_i.$$

The incidence graph between type-i vertices and cotype-i edges is therefore

$$\mathrm{CC}(G_s; K_i, U_i) = \mathrm{CC}(G_s; U_{\widehat{i}}, U_i),$$

which is the same bipartite graph as $\mathrm{CC}(G_s; U_i, U_{\widehat{i}})$ after swapping the two sides.

Thus the apparently global graph $Y_{i,s}$ has a simple intrinsic interpretation:

$$\boxed{Y_{i,s} = \text{incidence graph between type-}i\text{ vertices and cotype-}i\text{ edges in } X_s.}$$

This observation is what makes the swap-walk proof possible.

5 The local Heisenberg link calculation

We next compute the vertex links of X_s.

Fix a type-i vertex, say the identity coset K_i. Its link is the coset graph

$$\mathrm{CC}(K_i; K_i \cap K_j, K_i \cap K_k),$$

where $\{i, j, k\} = \{0, 1, 2\}$. By Lemma 3.1, this is the coset graph of a Heisenberg group with respect to two horizontal subgroups.

We now compute the spectrum of this graph explicitly.

Lemma 5.1 (Heisenberg incidence spectrum)**.** *Let*

$$H = \mathbb{F}_q^3$$

with multiplication

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy').$$

Let

$$X = \{(x, 0, 0) : x \in \mathbb{F}_q\}, \qquad Y = \{(0, y, 0) : y \in \mathbb{F}_q\}.$$

Then the normalized nontrivial singular value of $\mathrm{CC}(H; X, Y)$ *is exactly*

$$q^{-1/2}.$$

Consequently every vertex link of X_s has normalized nontrivial singular value $q^{-1/2}$.

Proof. Right X-cosets are parametrized by pairs $(y,z) \in \mathbb{F}_q^2$. Right Y-cosets are parametrized by pairs $(x,w) \in \mathbb{F}_q^2$, where the coset represented by $(x,0,w)$ consists of

$$(x, y, w + xy), \qquad y \in \mathbb{F}_q.$$

Thus the incidence relation is

$$(y,z) \sim (x,w) \quad \Longleftrightarrow \quad z = w + xy.$$

The normalized adjacency operator from the Y-side to the X-side is therefore

$$Af(y,z) = \frac{1}{q} \sum_{x \in \mathbb{F}_q} f(x, z - xy).$$

Fix a nontrivial additive character ψ of $\mathbb{F}_q$, and define

$$f_{\alpha,\beta}(x,w) = \psi(\alpha x + \beta w), \qquad \alpha, \beta \in \mathbb{F}_q.$$

Then

$$\begin{aligned} Af_{\alpha,\beta}(y,z) &= \frac{1}{q} \sum_{x \in \mathbb{F}_q} \psi(\alpha x + \beta(z - xy)) \\ &= \psi(\beta z) \frac{1}{q} \sum_{x \in \mathbb{F}_q} \psi((\alpha - \beta y)x). \end{aligned}$$

If $(\alpha,\beta) = (0,0)$, this is the constant singular vector with singular value 1. If $\beta = 0$ and $\alpha \neq 0$, the output is zero. If $\beta \neq 0$, the output is supported on the single horizontal fiber $y = \alpha/\beta$ and has normalized L^2-norm $q^{-1/2}$ times the normalized L^2-norm of the input.

These characters form an orthonormal basis of functions on the Y-side, so the largest nontrivial singular value is exactly $q^{-1/2}$. □

6 The colored swap-walk input

We use one standard local-to-global statement for partite high-dimensional expanders. In this note we only need the two-dimensional special case.

Proposition 6.1 (Colored swap-walk estimate, special case)**.** *Let X be a finite pure tripartite 2-complex, equipped with the probability measure induced by a uniformly random chamber. Suppose X is connected and its relevant two-color links have normalized nontrivial singular value at most λ. Then, for each color i, the incidence graph between type-i vertices and cotype-i edges has normalized nontrivial singular value at most*

$$2\lambda.$$

Equivalently, if M_i is the normalized operator from cotype-i edges to type-i vertices, then

$$\|M_i\|_{L_0^2 \to L_0^2} \leq 2\lambda.$$

This is the $|I| = 1$, $|J| = 2$ case of the colored swap-walk theorem. In the notation often used for colored swaps,

$$\lambda(S_{I,J}) \le |I|\,|J|\,\lambda.$$

See, for example, the colored swap theorem of Dikstein–Dinur and the formulation recorded in later treatments of swap walks and local spectral expansion. Related swap-walk estimates also appear in Alev–Jeronimo–Tulsiani's work on high-dimensional expanders and CSPs.

Remark 6.2 (What is hidden in Proposition 6.1)**.** The proposition is the only imported HDX tool in the proof. It packages the usual local-to-global mechanism: a walk that swaps a face of one color set against a disjoint face of another color set has its nontrivial spectrum controlled by the spectra of the relevant links. For this note, the factor 2 comes from swapping one color against two colors. If one chooses a different normalization of local spectral expansion, the displayed numerical constant may change, but the conclusion remains $O(q^{-1/2})$.

7 Proof of the codimension-one bound

We now prove Theorem 2.2.

Fix $i \in \{0, 1, 2\}$ and write

$$K_i = U_{\widehat{i}}.$$

By Section 4, the cotype-i edges of X_s are indexed by G_s/U_i, while the type-i vertices are indexed by G_s/K_i. Hence the colored incidence graph in Proposition 6.1 is precisely

$$\mathrm{CC}(G_s; K_i, U_i) = \mathrm{CC}(G_s; U_{\widehat{i}}, U_i).$$

This is the same graph as $Y_{i,s} = \mathrm{CC}(G_s; U_i, U_{\widehat{i}})$ after swapping sides.

By Lemma 5.1, every vertex link contributing to this colored walk has normalized nontrivial singular value

$$\lambda = q^{-1/2}.$$

Applying Proposition 6.1 gives

$$\sigma_2(Y_{i,s}) \le 2q^{-1/2}.$$

Finally, under the chamber identification $\ell^2(G_s)$, the normalized bipartite adjacency operator of $Y_{i,s}$ has the same nontrivial singular values as

$$P_{U_i} P_{K_i} = P_{U_i} P_{U_{\widehat{i}}}.$$

Constants account for the trivial singular value 1, and therefore

$$\left\| P_{U_i} P_{U_{\widehat{i}}} \right\|_{\ell_0^2(G_s) \to \ell_0^2(G_s)} \le 2q^{-1/2}.$$

This proves the theorem uniformly in s, q, and i.

8 What this result is, and what it is not

It is useful to be precise about the status of the theorem.

What the theorem gives

It gives a uniform spectral bound for the codimension-one incidence graph

$$\mathrm{CC}(G_s; U_i, U_{\widehat{i}}).$$

Since $|U_i| = q$ and $|U_{\widehat{i}}| = q^3$, this graph is (q, q^3)-biregular. The normalized bound

$$\sigma_2 \le 2q^{-1/2}$$

corresponds to the unnormalized singular-value bound

$$\lambda_2^{\mathrm{un}} \le 2q^2 \cdot q^{-1/2} = 2q^{3/2}.$$

Measured against the larger degree q^3, this is a $D^{1/2}$-scale estimate.

The result is also level-uniform: nothing in the proof depends on s beyond the condition $s \ge 2$.

What the theorem does not give

The theorem does not compute the reduced spectrum of an infinite KMS complex. It does not prove a sharp Ramanujan bound in the universal-cover sense. It also does not identify the optimal leading constant. Earlier direct Fourier calculations at level $s = 2$ suggest that for at least one side of the affine $\widetilde{A}_2$ model one has

$$\sigma_2^2 = \frac{2q^2 - q + 1}{q^3},$$

so the leading constant may be $\sqrt{2}$ rather than 1 or 2. That exact calculation is not used in the present proof.

Most importantly, the proof shows that this particular codimension-one shadow is controlled by standard colored swap-walk technology. Therefore, as a coarse $O(q^{-1/2})$ theorem, it is best viewed as a consequence of local spectral expansion rather than as a new global Ramanujan phenomenon.

9 Consequences and nearby graph shadows

9.1 Layered or gallery shadows

Once the one-coordinate bound is available, any layered product of such shadows contracts multiplicatively on ℓ_0^2. For example, given a word

$$w = (i_1, \dots, i_\Delta),$$

consider

$$T_w = (P_{U_{i_\Delta}} P_{U_{\widehat{i_\Delta}}}) \cdots (P_{U_{i_1}} P_{U_{\widehat{i_1}}}).$$

Then

$$\|T_w\|_{\ell_0^2 \to \ell_0^2} \le (2q^{-1/2})^\Delta.$$

Combinatorially, this is the normalized adjacency operator of a layered multigraph whose edges are length-Δ galleries built from codimension-one shadows.

This observation may be relevant if one wants distance-sensitive spectral decay. It is not the same as proving that a simultaneous graph

$$\mathrm{CC}(G_s; U_D, U_{\widehat{D}})$$

has full spectrum $q^{-|D|/2}$. Such simultaneous graphs generally have lower-coordinate eigenfunctions, so the correct higher-rank statement would need a "top-new" projection that removes lower-residue functions.

9.2 Balanced chamber walks

The balanced chamber operator

$$W = P_{U_2} P_{U_1} P_{U_0}$$

corresponds to the bipartite chamber graph

$$g_L \sim (g u_0 u_1 u_2)_R, \qquad u_i \in U_i.$$

This graph has degree q^3. The codimension-one theorem can be used as an input for bounding such balanced gallery operators, though the cleanest bounds depend on how one groups the product into rank-two or codimension-one pieces. In any case, the same philosophy applies: local Heisenberg expansion controls the large chamber-level incidence operators once they are expressed as colored swaps or products of colored swaps.

10 Possible HLMRZ-style applications

This section is deliberately forward-looking. It records how the KMS-shadow viewpoint might be used in constructions inspired by Hsieh–Lubotzky–Mohanty–Reiner–Zhang (HLMRZ), without claiming that the present theorem by itself supplies all needed ingredients.

10.1 What HLMRZ needs from its base graphs

HLMRZ construct explicit lossless vertex expanders using the tripartite line product. The construction starts with a tripartite base graph

$$L \; -\!- \; M \; -\!- \; R$$

and places a constant-sized biregular gadget at every middle vertex $m \in M$. In HLMRZ, the base graphs come from coded face–vertex incidence graphs in Ramanujan Cayley cubical complexes. The cubical input is not merely spectral. It provides several forms of structure:

(i) **Spectral/skeleton expansion.** The relevant base and skeleton graphs must have strong spectral control.

(ii) **Small-set high-neighbor control.** For a small subset of one side, not too many middle vertices should have abnormally many neighbors in that subset. In the cubical case this is proved via subcube-density or Loomis–Whitney-type estimates.

(iii) **Common-neighborhood structure.** Collisions in the tripartite product are controlled using detailed information about common neighborhoods, not just their average size.

(iv) **Canonical labels.** At each middle vertex, the left and right neighbors must be ordered or labeled coherently so that the same fixed gadget can be placed in a structured way.

(v) **Symmetry.** For some downstream applications, it is useful that the final graph admits a free group action inherited from the base geometry.

The present KMS shadow theorem addresses only the first item for a particular family of incidence graphs. The other items remain substantial work.

10.2 A candidate KMS replacement for cubical incidence

The natural KMS analogue of a cubical face–vertex incidence graph would be a residue-incidence graph in a higher-rank KMS coset complex. Schematically, one might take

$$X = \mathrm{CC}(G; (K_i)_{i \in I})$$

for a finite KMS quotient G, and choose families of residues

$$\mathcal{L}, \qquad \mathcal{M}, \qquad \mathcal{R}$$

so that incidence gives bipartite graphs

$$\mathcal{L} \; - - \; \mathcal{M}, \qquad \mathcal{M} \; - - \; \mathcal{R}.$$

The codimension-one graphs in this note are the rank-three shadows where one side is a vertex type and the other side is an opposite edge type.

For an HLMRZ-type line product, the ideal KMS base would satisfy:

K1. The incidence graphs $\mathcal{L} - -\mathcal{M}$ and $\mathcal{M} - -\mathcal{R}$ have controlled nontrivial singular values, preferably from colored swap-walk bounds.

K2. The skeleton graph on $\mathcal{M}$ generated by two-step paths has good small-set spectral or combinatorial expansion.

K3. Common neighborhoods of pairs of middle vertices decompose into uniformly described algebraic fibers, ideally expressible in root-subgroup coordinates.

K4. Small subsets of $\mathcal{M}$ satisfy KMS analogues of cubical subcube-density bounds: few residues should be very rich in a small set.

K5. The neighbor labels at each middle vertex come from a normal form in the relevant root groups, giving canonical gadget placement.

Items K1 and some cases of K2 look accessible with the existing local spectral expansion and colored swap-walk theory. Items K3 and K4 are the hard combinatorial replacements for the cubical machinery.

10.3 Why KMS geometry is tempting

KMS/coset constructions have several attractive features for this purpose.

First, the complexes are highly explicit: vertices and faces are cosets, and incidence is group-theoretic containment. Second, chambers are often indexed by group elements, giving a large free or nearly free group action. Third, local links are finite algebraic incidence geometries, frequently Heisenberg or Chevalley-type graphs, whose spectra are computable. Fourth, root-subgroup normal forms may provide the canonical labels needed for placing a constant-sized gadget.

The main caveat is that KMS coordinates are noncommutative. Cubes have product coordinates and Loomis–Whitney inequalities essentially for free. In KMS geometries, commutators introduce additional root coordinates. This does not make the program impossible, but it means that the analogues of subcube-density and common-neighborhood lemmas must be proved in triangular root-coordinate systems rather than in literal product spaces.

11 Side observations not used in the proof

1. **Exact level-two spectra.** In the level-$s = 2$ affine $\widetilde{A}_2$ model, squaring $\mathrm{CC}(G_s; U_i, U_{\widehat{i}})$ can reduce to an abelian Cayley walk on a vector space. For one orientation, a direct point-count gives
$$\sigma_2^2 = \frac{2q^2 - q + 1}{q^3}.$$
This suggests the sharp leading constant may be $\sqrt{2}$ for the normalized singular value. The swap-walk proof gives the coarser constant 2.

2. **Universal-cover interpretation.** A stronger "Ramanujan KMS" statement would compare finite quotient spectra with the reduced spectrum of the corresponding operator on the infinite KMS group. The present theorem does not identify that reduced norm.

3. **Higher-rank top-new spectrum.** For simultaneous residue graphs $\mathrm{CC}(G; U_D, U_{\widehat{D}})$, one should not expect full-spectrum decay $q^{-|D|/2}$ because lower-coordinate eigenfunctions survive. The plausible statement is a top-new bound after removing lower-residue subspaces.

4. **Small q.** When $q \leq 4$, the displayed bound $2q^{-1/2}$ is at least 1, hence trivial as a bound on singular values. The theorem is asymptotic in spirit.

12 Open directions

The following list is ranked by a mixture of mathematical importance and perceived probability of success.

Rank	Direction	Importance	Probability
1	Prove sharp constants for $\mathrm{CC}(G_s; U_i, U_{\widehat{i}})$, and identify whether the reduced/infinite-cover value is $\sqrt{2/q} + O(q^{-1})$ or something nearby.	High	Medium
2	Formulate and prove a top-new theorem for simultaneous higher-rank shadows $\mathrm{CC}(G; U_D, U_{\widehat{D}})$ after removing all lower-residue spectrum.	Very high	Medium-low
3	Develop KMS small-set density estimates: analogues of cubical subcube-density or triangle-density bounds for residue-incidence systems.	Very high	Low-medium
4	Prove structured common-neighborhood decompositions in root coordinates, suitable for line-product gadget placement.	High	Medium
5	Generalize the codimension-one proof to broader KMS diagrams and record explicit constants in terms of rank-two local geometries.	Medium-high	High for coarse bounds
6	Run systematic numerical experiments for rank 4 and beyond, comparing swap-walk predictions, exact spectra where computable, and small-set rich-residue counts.	Medium	High

The most conceptually important next step is not another proof of the coarse $O(q^{-1/2})$ bound. That is now explained by colored swap walks. The next genuinely new step is either the sharp spectral problem or the KMS small-set combinatorics needed to replace cubical incidence systems in HLMRZ-type arguments.

Acknowledgment of proof status

The proof in this note should be read as a clean repackaging of standard local-to-global HDX technology applied to a useful coset-incidence graph. Its value is mainly organizational: it identifies the codimension-one graph as a colored swap graph inside a tripartite KMS coset complex and isolates the remaining questions that are not settled by trickling down.

References

[1] V. L. Alev, F. G. Jeronimo, and M. Tulsiani. Approximating constraint satisfaction problems on high-dimensional expanders. In *FOCS 2019*, pp. 180–201, 2019.

[2] Y. Dikstein and I. Dinur. Agreement/swap-walk theorems for high-dimensional expanders; see also later formulations in swap cosystolic expansion. Related exposition: *Swap Cosystolic Expansion*, arXiv:2312.15325.

[3] J.-T. Hsieh, T.-C. Lin, S. Mohanty, R. O'Donnell, and R. Y. Zhang. Explicit two-sided vertex expanders beyond the spectral barrier. arXiv:2411.11627, 2024.

[4] J.-T. Hsieh, A. Lubotzky, S. Mohanty, A. Reiner, and R. Y. Zhang. Explicit lossless vertex expanders. arXiv:2504.15087, 2025.

[5] T. Kaufman and I. Oppenheim. High order random walks: beyond spectral gap. *Combinatorica* 40, 245–281, 2020. Conference version: APPROX/RANDOM 2018.

[6] L. Grave de Peralta and I. Valentiner-Branth. High-dimensional expanders from Kac–Moody–Steinberg groups. arXiv:2401.05197, 2024/2025.

[7] R. O'Donnell and K. Pratt. High-dimensional expanders from Chevalley groups. *37th Computational Complexity Conference (CCC 2022)*, LIPIcs 234, Article 18, 2022.